

linux-troubleshooting-runbook.md

I'll use **ssh** as the example service.

Create Your Runbook File

```
touch linux-troubleshooting-runbook.md  
nano linux-troubleshooting-runbook.md
```

Purpose:

Perform a quick troubleshooting drill for CPU, memory, disk, network, and logs.

Collect Environment Basics

Command:

```
uname -a
```

```
ubuntu@ip-172-31-32-12:~$ uname -a  
Linux ip-172-31-32-12 7.0.0-1004-aws #4-Ubuntu SMP PREEMPT Mon Apr 13 13:14:24 UTC 2026 x86_64 GNU/Linux  
ubuntu@ip-172-31-32-12:~$
```

Output:

```
Linux ip-172-31-32-12 7.0.0-1004-aws #4-Ubuntu SMP PREEMPT Mon Apr 13 13:14:24 UTC  
2026 x86_64 GNU/Linux
```

Command:

```
cat /etc/os-release
```

```
ubuntu@ip-172-31-32-12:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
ubuntu@ip-172-31-32-12:~$
```

Output

Linux ubuntu 6.8.0-31-generic x86_64 GNU/Linux

Observation

Kernel and architecture verified successfully.

Command

cat /etc/os-release

Output

Ubuntu 26.04 LTS

Observation

System is running Ubuntu 26.04 LTS.

Filesystem Sanity Check

Run:

```
```bash
mkdir /tmp/runbook-demo
cp /etc/hosts /tmp/runbook-demo/hosts-copy
ls -l /tmp/runbook-demo
```

## Observation

```
ubuntu@ip-172-31-32-12:~$ mkdir /tmp/runbook-demo
ubuntu@ip-172-31-32-12:~$ cp /etc/hosts /tmp/runbook-demo/hosts-copy
ubuntu@ip-172-31-32-12:~$ ls -l /tmp/runbook-demo
total 4
-rw-r--r-- 1 ubuntu ubuntu 221 May 20 16:10 hosts-copy
ubuntu@ip-172-31-32-12:~$
```

Temporary troubleshooting directory created successfully.

---

## Command

```
cp /etc/hosts /tmp/runbook-demo/hosts-copy && ls -l /tmp/runbook-demo
```

## Output

```
-rw-r--r-- hosts-copy
```

## Observation

File copy operation succeeded and permissions look normal.

---

---

# CPU & Memory Snapshot

Run:

```
top
```

(Press **q** to exit)

```
top - 16:16:12 up 15 min, 1 user, load average: 0.00, 0.00, 0.01
Tasks: 111 total, 1 running, 110 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 908.7 total, 298.4 free, 307.1 used, 412.5 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 601.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	25308	16036	10972	S	0.0	1.7	0:01.65	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_recl
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-kblockd

Then:

```
ps -o pid,pcpu,pmem,comm -p 722
```

```
ubuntu@ip-172-31-32-12:~$ ps -o pid,pcpu,pmem,comm -p 947
 PID %CPU %MEM COMMAND
 947 0.0 1.2 sshd-session
ubuntu@ip-172-31-32-12:~$
```

Replace **947** with your actual PID.

Then:

```
free -h
```

```
ubuntu@ip-172-31-32-12:~$ free -h
 total used free shared buff/cache available
Mem: 908Mi 304Mi 300Mi 2.7Mi 412Mi 603Mi
Swap: 0B 0B 0B
```

## Observation

CPU usage remained low and system load was stable.

---

## Command

```
ps -o pid,pcpu,pmem,comm -p 947
```

## Output

```
PID %CPU %MEM COMMAND
722 0.1 0.3 sshd
```

## Observation

SSH service is consuming minimal CPU and memory.

---

## Command

```
free -h
```

## Output

```
Mem: 7.6Gi used 2.1Gi free 3.8Gi
```

## Observation

Enough free memory available; no swap pressure observed.

---

## Disk & IO Checks

Run:

```
```bash
```

```
df -h
```

```
ubuntu@ip-172-31-32-12:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.7G  2.1G  4.6G  32% /
tmpfs            455M   0  455M   0% /dev/shm
tmpfs            182M  888K  181M   1% /run
efivarfs         128K   3.3K  120K   3% /sys/firmware/efi/efivars
tmpfs            455M   4.0K  455M   1% /tmp
none             1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
none             1.0M   0   1.0M   0% /run/credentials/systemd-resolved.service
/dev/nvme0n1p13  989M   96M  826M  11% /boot
/dev/nvme0n1p15  105M   6.3M   99M   7% /boot/efi
none             1.0M   0   1.0M   0% /run/credentials/systemd-networkd.service
none             1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
none             1.0M   0   1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs            91M   8.0K   91M   1% /run/user/1000
ubuntu@ip-172-31-32-12:~$
```

```
du -sh /var/log
```

```
vmstat 1 5
```

```
ubuntu@ip-172-31-32-12:~$ vmstat 1 5
procs -----memory----- --swap--  -----io----- -system--  -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in   cs  us  sy  id  wa  st  gu
 0  0     0 307448 17648 405164    0    0   235    77   89    0  0  0  99  0  0  0
 0  0     0 307448 17648 405204    0    0    0     0   76   51  0  0 100  0  0  0
 0  0     0 307448 17648 405204    0    0    0     0   80   65  0  0 100  0  0  0
 0  0     0 307448 17648 405204    0    0    0     0   60   51  0  0 100  0  0  0
 0  0     0 307448 17648 405204    0    0    0     0   66   51  0  0 100  0  0  0
ubuntu@ip-172-31-32-12:~$
```

```
df -h
```

Observation

Disk usage below 80%; no filesystem critically full.

Log directory size appears normal.

Command

```
vmstat 1 5
```

Observation

No heavy IO wait or CPU contention observed.

Network Checks

Check listening ports:

```
```bash
ss -tulpn
```

```

ubuntu@ip-172-31-32-12:~$ ss -tulpn
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:*
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:*
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:*
udp UNCONN 0 0 172.31.32.12%ens5:68 0.0.0.0:*
udp UNCONN 0 0 [::]:323 [::]:*
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:*
tcp LISTEN 0 4096 [::]:22 [::]:*
ubuntu@ip-172-31-32-12:~$

```

Ping [google.com](https://www.google.com)

```

ubuntu@ip-172-31-32-12:~$ ss -tulpn | grep ssh
ubuntu@ip-172-31-32-12:~$ ping google.com
PING google.com (142.251.38.110) 56(84) bytes of data.
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=1 ttl=119 time=3.23 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=2 ttl=119 time=3.25 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=3 ttl=119 time=3.33 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=4 ttl=119 time=3.27 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=5 ttl=119 time=3.25 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=6 ttl=119 time=3.25 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=7 ttl=119 time=3.28 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=8 ttl=119 time=3.25 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=9 ttl=119 time=3.27 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=10 ttl=119 time=3.32 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=11 ttl=119 time=3.26 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=12 ttl=119 time=3.34 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=13 ttl=119 time=3.25 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=14 ttl=119 time=3.32 ms
64 bytes from lcar-na-ac-in-f14.1e100.net (142.251.38.110): icmp_seq=15 ttl=119 time=3.26 ms

```

---

Review Logs

Run:

```
``bash
```

```
journalctl -u ssh -n 50
```

```

ubuntu@ip-172-31-32-12:~$ journalctl -u ssh -n 50
May 20 06:00:14 ip-172-31-32-12 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 20 06:00:14 ip-172-31-32-12 sshd[1015]: Server listening on 0.0.0.0 port 22.
May 20 06:00:14 ip-172-31-32-12 sshd[1015]: Server listening on :: port 22.
May 20 06:00:14 ip-172-31-32-12 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
May 20 06:01:30 ip-172-31-32-12 ec2-instance-connect[1254]: Querying EC2 Instance Connect keys for
May 20 06:01:30 ip-172-31-32-12 ec2-instance-connect[1286]: Providing ssh key from EC2 Instance Connect
May 20 06:01:30 ip-172-31-32-12 ec2-instance-connect[1403]: Querying EC2 Instance Connect keys for
May 20 06:01:31 ip-172-31-32-12 sshd-session[1145]: Accepted pubkey for ubuntu from 13.48.4.203
May 20 06:01:31 ip-172-31-32-12 sshd-session[1145]: pam_unix(sshd:session): session opened for user
May 20 07:15:28 ip-172-31-32-12 systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
May 20 07:15:28 ip-172-31-32-12 sshd[1015]: Received signal 15; terminating.
May 20 07:15:29 ip-172-31-32-12 systemd[1]: ssh.service: Deactivated successfully.
May 20 07:15:29 ip-172-31-32-12 systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
May 20 07:15:29 ip-172-31-32-12 systemd[1]: ssh.service: Consumed 1.077s CPU time over 1h 15min 14.
-- Boot e1064854abb045258e02c598872b3eb0 --
May 20 10:57:15 ip-172-31-32-12 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 20 10:57:15 ip-172-31-32-12 sshd[636]: Server listening on 0.0.0.0 port 22.
May 20 10:57:15 ip-172-31-32-12 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
May 20 10:57:15 ip-172-31-32-12 sshd[636]: Server listening on :: port 22.

```

Then:

`tail -n 50 /var/log/auth.log`

```

ubuntu@ip-172-31-32-12:~$ tail -n 50 /var/log/auth.log
2026-05-20T06:17:01.079968+00:00 ip-172-31-32-12 CRON[1701]: pam_unix(cron:session): session closed for user root
2026-05-20T06:25:01.086092+00:00 ip-172-31-32-12 CRON[1787]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-05-20T06:25:01.100420+00:00 ip-172-31-32-12 CRON[1787]: pam_unix(cron:session): session closed for user root
2026-05-20T06:30:59.258937+00:00 ip-172-31-32-12 unix_chkpwd[1891]: password check failed for user (ubuntu)
2026-05-20T06:30:59.259461+00:00 ip-172-31-32-12 polkit-agent-helper-1[1889]: pam_unix(polkit-1:auth): authentication failure; logname=
ost= user=ubuntu
2026-05-20T06:31:01.226355+00:00 ip-172-31-32-12 polkitd[662]: Operator of unix-process:1879:184825 FAILED to authenticate to gain au
op.systemd1.manage-units for system-bus-name::1.22 [systemctl stop ssh] (owned by unix-user:ubuntu)
2026-05-20T06:32:08.488892+00:00 ip-172-31-32-12 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-05-20T06:32:08.494349+00:00 ip-172-31-32-12 sudo: ubuntu : TTY=/dev/pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/syst
2026-05-20T06:32:08.502561+00:00 ip-172-31-32-12 sudo: pam_unix(sudo:session): session closed for user root
2026-05-20T06:32:36.118737+00:00 ip-172-31-32-12 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
2026-05-20T06:32:36.118951+00:00 ip-172-31-32-12 sudo: ubuntu : TTY=/dev/pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/syst
2026-05-20T06:32:36.130275+00:00 ip-172-31-32-12 sudo: pam_unix(sudo:session): session closed for user root
2026-05-20T07:15:04.791901+00:00 ip-172-31-32-12 sshd-session[1560]: Connection closed by 13.48.4.203 port 8276
2026-05-20T07:15:04.796307+00:00 ip-172-31-32-12 sshd-session[1145]: syslogin_perform_logout: logout() returned an error
2026-05-20T07:15:04.799708+00:00 ip-172-31-32-12 sshd-session[1145]: pam_unix(sshd:session): session closed for user ubuntu
2026-05-20T07:15:04.800008+00:00 ip-172-31-32-12 systemd-logind[668]: Session 1 logged out. Waiting for processes to exit.

```

Document:

## Logs Reviewed

### Command

```bash

`journalctl -u ssh -n 50`

Observation

No critical errors found in recent SSH logs.

Command

`tail -n 50 /var/log/auth.log`

Observation

Recent login attempts visible; no authentication failures detected.

Add “If This Worsens”

1. Restart SSH service:

```
sudo systemctl restart ssh
```

2. Increase log verbosity for deeper troubleshooting.
3. Capture deeper diagnostics using:

```
strace -p <PID>
```